PulseIQ Analytics

PHASE 1 REPORT | FOUNDATION | 24 OCTOBER 2026

Participant	Arpita Madpalli	Track	Data Analytics
Repository	https://github.com/ArpitaM13/pulseiq Public; fobesitech listed as collaborator	Tag	phase-1
Latest commit	355fa42 - Exclude temporary editor backups	Tests	20 passed in 40.31 seconds

Project summary

PulseIQ Analytics is a local workforce and business analytics project using a fictional company workbook. The pipeline ingests 33 worksheets into DuckDB, cleans five core workforce tables, builds warehouse and analytics layers, validates supplied contracts, runs 14 data-quality checks, and serves governed metric views to a multi-page Streamlit dashboard.

33	20	28	14
Workbook sheets loaded	Automated tests passed	Metric definitions	DQ checks executed

Phase 1 task status

Task	Implementation and evidence	Status
1. ETL and workspace	33 sheets loaded; all-worksheet row audit reports conservation and raw-to-warehouse count match. Five core tables cleaned, two reversed-date leave rows quarantined. Contract validation still has 1 FAIL and 10 WARN.	**Implemented; findings retained**
2. Star-schema warehouse	Warehouse/staging layers build. Schema, primary-key and foreign-key/orphan audits are generated; known orphan records remain documented for review.	**Implemented; audit reviewed**
3. Quality gatekeeper	All 14 catalogue checks execute; known-issue review generated. Results are 4 FAIL, 7 WARN and 3 PASS.	**Checks execute; issues surfaced**
4. Governed metrics	28 metric definitions/views; independent checks cover 3,069 rows with zero failures. Small-group suppression and catalog/lineage support are included.	**Implemented; spot-checks pass**
5. Workforce dashboard	KPI cards, single visible-total headcount trend, department mix, work-mode and city-tier charts. Filter-key and chart-label refinements applied; test filters before submission.	**Visuals verified; final interaction check**

Pipeline flow: Excel workbook -> raw DuckDB -> ETL/quarantine -> warehouse + audits -> governed metrics -> Streamlit dashboard. Source defects are reported, not silently corrected.

Engineering evidence and quality findings

The latest shared local pipeline run completed all six stages on 23 October 2026. The following counts and statuses come from the terminal output and generated audit reports.

Core ETL row accounting

Core table	Rows in	Accepted	Quarantined
dim_employee	1,606	1,606	0
fact_payroll	7,344	7,344	0
fact_attendance_monthly	7,343	7,343	0
fact_leave	3,441	3,439	2
fact_headcount_monthly	891	891	0
TOTAL	20,625	20,623	2

The two quarantined leave records (LR-000025 and LR-002286) have start dates after end dates and remain in rejected outputs with an explicit reason. The 33-worksheet row audit reports all rows conserved and raw-to-warehouse counts matched.

Contract and quality-gate results

Validation item	Current result	Interpretation
Supplied column contracts	51 rule/column rows; 1 FAIL, 10 WARN	Unexpected columns are warned and preserved; one completed_date validation failure remains. Review before sign-off.
14 DQ checks	4 FAIL, 7 WARN, 3 PASS	Known training defects remain visible: payroll/ledger mismatch, duplicate payroll rows, orphan employee keys, and one history overlap.
Metric spot-checks	3,069 rows; 0 failures	Independent pandas calculations matched the tested governed SQL outputs.
Automated tests	20 passed in 40.31 seconds	Latest reported local test run; rerun after any further code edits.

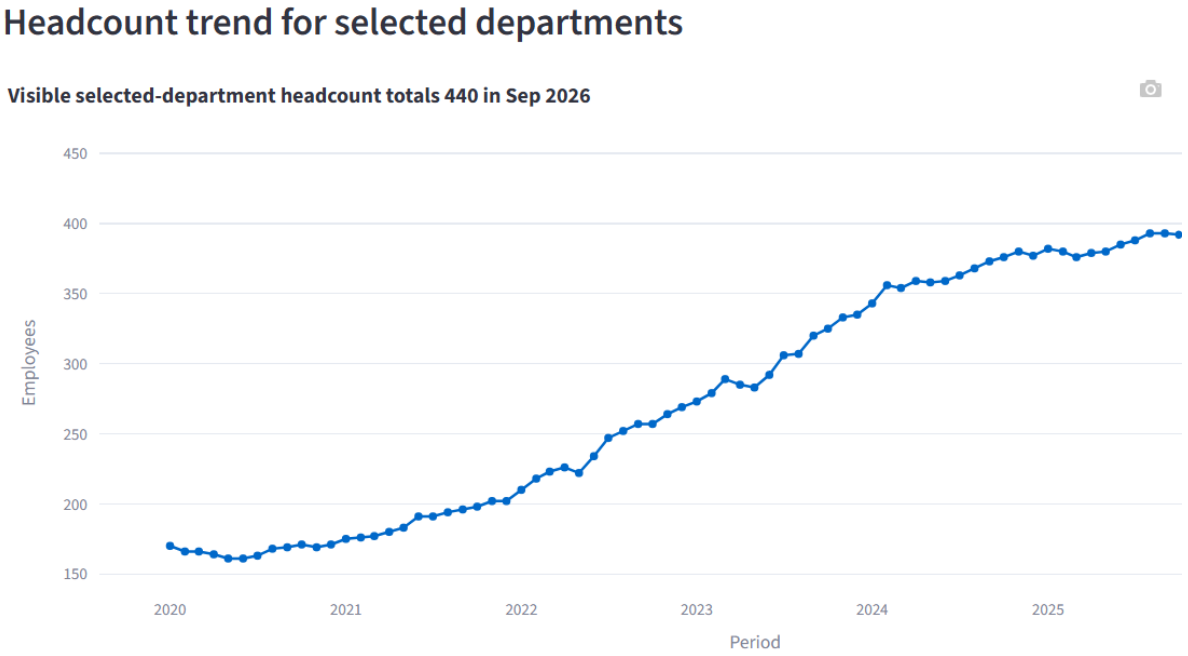
Observed DQ checks

Check	Status	Affected	Check	Status	Affected
DQ-C01	FAIL	5	**DQ-C08**	WARN	1
DQ-C02	PASS	0	**DQ-C09**	WARN	20
DQ-C03	FAIL	2	**DQ-C10**	WARN	2
DQ-C04	FAIL	2	**DQ-C11**	WARN	5
DQ-C05	WARN	4	**DQ-C12**	FAIL	1
DQ-C06	WARN	1	**DQ-C13**	PASS	0
DQ-C07	WARN	1	**DQ-C14**	PASS	0

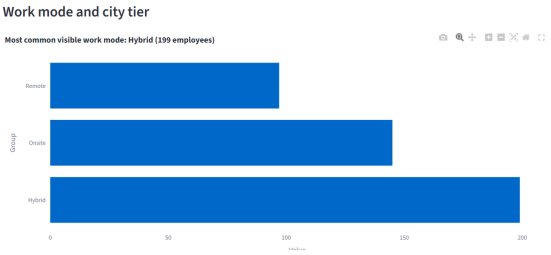
Audit evidence: reports/table_row_audit.csv, warehouse_row_counts.csv, primary_key_audit.csv, foreign_key_audit.csv, contract_validation.csv, dq_results.csv, known_issue_review.csv and manual_metric_checks.csv. Critical defects remain open for investigation; they are not production sign-off.

Dashboard evidence, learning and next steps

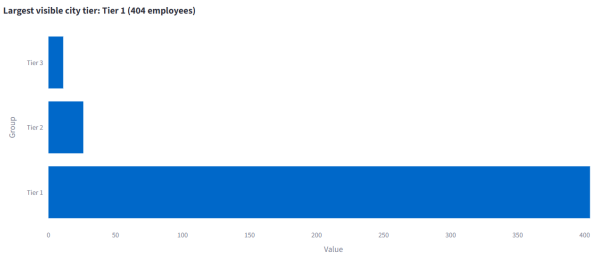
Task 5 - Workforce & Headcount | Evidence captured from the local Streamlit app



Headcount trend: the visible selected-department total grew by 270 from January 2020 to September 2026, reaching 440 visible employees in September 2026. The company-wide KPI is 441 because small-group suppression can remove a group from the detailed chart.



Work mode: Hybrid is the largest visible group (199 employees).



City tier: Tier 1 is the largest visible tier (404 employees).

What I learned

I learned to ingest workbook data into DuckDB, preserve raw records, create clean and quarantined outputs, and reconcile row counts. I learned why data contracts and quality checks make an ETL pipeline auditable, how governed SQL views keep metric definitions consistent, and how to validate selected metrics independently with pandas. I also learned to use Streamlit and Plotly to present workforce trends with privacy-aware group suppression.

Problems solved

Fixed the composite-key validation that incorrectly rejected valid headcount rows; resolved DuckDB SQL alias/syntax problems; added unique Streamlit filter keys; and simplified the headcount chart to one visible-total trend line. Source-data findings (including payroll reconciliation, duplicate/orphan payroll rows, reversed leave dates and employee-history overlap) remain visible for audit.

Plan for Phase 2

Extend attendance, attrition and team-productivity analysis; investigate payroll-to-ledger differences against the source definitions; document actions and ownership for open quality findings; and keep the pipeline reproducible with automated tests.

GITHUB REPOSITORY EVIDENCE

Commit: 355fa42 - Exclude temporary editor backups | Tag: phase-1 | Working tree: clean | Last test run: 20 passed in 40.31 seconds.

Repository: https://github.com/ArpitaM13/pulseiq (public). Phase 1 commit and tag pushed successfully.
GitHub access settings show fobesitech listed as a collaborator.